

Hypothesis Testing Summary

Objective

To determine whether the average customer spending differs significantly across different age groups.

Hypotheses

- **H₀:** There is no significant difference in average customer spending among Gen Z, Millennials, and Gen X.
- **H₁:** There is a significant difference in average customer spending among Gen Z, Millennials, and Gen X.

Test Applied

One-Way ANOVA

Test Results

Parameter	Result
Test	One-Way ANOVA
F-Statistic	0.0372
P-Value	0.9635
Significance Level	0.05

Decision

Since **0.9635 > 0.05**, the null hypothesis was **not rejected**.

Observation

```
===== HYPOTHESIS TEST =====  
F-Statistic : 0.0372  
P-Value    : 0.9635  
  
Result:  
Fail to Reject the Null Hypothesis  
No significant difference in average spending across age groups.
```

The dataset showed that **Millennials generated the highest total revenue** because they placed **more orders (442)** than the other age groups, **not because they spent significantly more per order**. That's exactly why the ANOVA test found **no significant difference in average spending**. These two findings are consistent with each other, and

mentioning this connection during your presentation will demonstrate a strong understanding of the data.

Conclusion

The analysis found **no statistically significant difference** in average customer spending across the three age groups. This suggests that spending behavior is relatively consistent regardless of age group, and businesses should consider additional customer attributes—such as purchase history, customer segment, or product preferences—for more effective marketing and customer engagement strategies.